AP PRECALCULUS

Unit 1 • Lesson 1.4 practice

Lesson 1.4 • Polynomial Functions and Rates of Change

5 multiple-choice questions and 1 free-response question

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

What is the greatest possible number of local extrema of a polynomial of degree 5?

(A) 5
(B) 4
(C) 3
(D) 6

Question 2

NO CALCULATOR

A polynomial p is increasing on $(-\infty, -2)$, decreasing on $(-2, 3)$ and increasing on $(3, \infty)$. Where does p have a local maximum?

(A) $x = 3$
(B) $x = 0$
(C) $x = -2$
(D) Nowhere

Question 3

NO CALCULATOR

What is the leading coefficient of $p(x) = 2x^3(x-1)^2 - x^5$?

(A) 2
(B) -1
(C) 1
(D) 3

Question 4

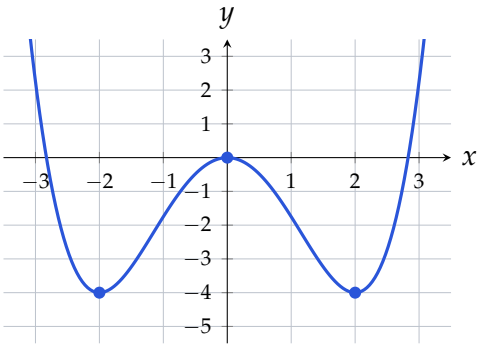
NO CALCULATOR

A point of inflection on the graph of a polynomial occurs where

(A) the rate of change changes from increasing to decreasing, or the reverse.
(B) the function changes from increasing to decreasing.
(C) the graph crosses the x-axis.
(D) the function has a local minimum.

Question 5 (graph) NO CALCULATOR

The graph of the polynomial $p(x) = \frac{1}{4}x^4 - 2x^2$ is shown. What is the absolute minimum value of p?



(A) 0
(B) -2
(C) -5
(D) -4

Question 6 (Free response) NO CALCULATOR

The polynomial $p(x) = x^3 - 6x^2 + 9x$ has a local maximum at $(1, 4)$, a local minimum at $(3, 0)$ and a point of inflection at $(2, 2)$.

(a) On which intervals is p increasing? decreasing?
(b) Find the absolute maximum and minimum values of p on $[0, 4]$.
(c) On which intervals is the graph of p concave up? concave down?